

Activity - 1

Stack, Queue and Linked List:

Stack and Queue:

Stack operations:

- * Push (54) $\rightarrow$ [54]
- * Push [52] $\rightarrow$ [54, 52]
- * Pop() $\rightarrow$ removes 52
- * push [55] $\rightarrow$ [54, 55]
- * push (62) $\rightarrow$ [54, 55, 62]
- * $s = \text{pop}()$ $\rightarrow$ 62

So, $s = 62$

Queue Operations:

- * enqueue (21) = [21]
- * enqueue (24) $\rightarrow$ [21, 24]
- * dequeue [] $\rightarrow$ removes 21
- * enqueue (28) $\rightarrow$ [24, 28]
- * enqueue (32) $\rightarrow$ [24, 28, 32]
- * $q = \text{dequeue}()$ $\rightarrow$ 24

So, $q = 24$

Answer :

$$s + q = 62 + 24 = 86$$

Postfix Expression :

$$105 + 606 / * 8 -$$

Evaluation :

$$* 105 + = 15$$

$$* 606 / = 10$$

$$* 15 \times 10 = 150$$

$$* 150 - 8 = 142$$

Answer :

142

3. The queue is reversed by removing elements from the front and inserting them back using enqueue operation

* The minimum number of enqueue operation required to reverse the given queue is 4

Answer = 4

4. The reverse operation can be simulation using only push and pop operation

The stack element are popped and pushed back in reverse order.

5. Expression : $8, 2, 3^* / 2, 3^* + 5, 1^* -$

$$2^3 = 8$$

$$\text{stack} = 8 / 8$$

$$8 / 8 = 1$$

$$\text{stack} = 1$$

$$2 * 3 = 6$$

$$\text{stack} = 1, 6$$

$$5 * 1 = 5$$

$$\text{stack} : 7, 5$$

$$7 - 5 = 2$$

$$\text{stack} = 2$$

Final Answer = 2

6. Infix : $a + b * c - d^e^f$

$$b * c = bc$$

$$a + bc^e = abc^e +$$

$$e^f = ef^e$$

$$d^e^f = def^{ee}$$

$$abc^e + - def^{ee} = abc^e + def^{ee}$$

Post fix $abc^e + def^{ee}$

7. Operation : push (10), push (20), pop, push (10), push (20), pop, push (40)

pop.

push (10) $\rightarrow$ [10]

push (20) $\rightarrow$ [10, 20]

pop $\rightarrow$ 20

push (10) $\rightarrow$ [10, 10]

push (20) $\rightarrow$ [10, 10, 20]

pop $\rightarrow$ 20

pop $\rightarrow$ 10

pop $\rightarrow$ 10

push (20) $\rightarrow$ 20

pop $\rightarrow$ 20

Values popped in order : 20, 20, 10, 10, 20

Values pushed in order : 10, 20, 10, 20